

LESSON 9.2

ADDING LINEAR EXPRESSIONS WITH RATIONAL COEFFICIENTS



LESSON INTRODUCTION

MA.7.AR.1.1 Apply properties of operations to add and subtract linear expressions with rational coefficients.

LEARNING TARGETS

- I can add linear expressions with rational coefficients.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.1.4 Apply the properties of operations to generate equivalent algebraic expressions with integer coefficients.

WORKING TOWARD

MA.8.AR.1.2 Apply properties of operations to multiply two linear expressions with rational coefficients.

ESSENTIAL QUESTIONS

- How can you represent algebraic expressions using algebra tiles?
- How is adding linear expressions algebraically similar to adding linear expressions using algebra tiles?

LESSON NARRATIVE

This lesson builds on Lesson 1 with a focus on adding linear expressions with rational numbers and exploring the effect of parentheses when adding linear expressions with algebra tiles. Students deepen their understanding of combining like terms and find the sum of algebraic expressions using the fewest possible terms.

COMPONENT SUMMARY

9.2.1 WARM-UP (~5 MIN.)

Explore, describe, and make predictions based on a visual pattern of squares

9.2.3 COLLABORATION (10 MIN.)

Students add linear expressions using algebra tiles

9.2.5 COLLABORATION (10 MIN.)

Work with a partner to find the sum of linear expressions algebraically using the fewest terms possible

9.2.7 WRAP-UP (~5 MIN.)

Error analysis for adding expressions

9.2.2 EXPLORATION (5 MIN.)

Consider four expressions and explain which one does not belong using math reasoning

9.2.4 GUIDED INSTRUCTION (10 MIN.)

Teacher models how to find the sum of linear expressions using the fewest terms possible without using algebra tiles

9.2.6 YOUR TURN! (5 MIN.)

Find the sum of linear expressions using the fewest terms possible

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Coefficient

MATERIALS

- Algebra tiles
- (Optional) Notecards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- If using concrete algebra tiles, determine how many tiles are needed to model all of the examples in the lesson and organize student sets in a container or Ziplock bag.
- If using virtual algebra tiles, practice using the manipulative and create student-friendly directions for how to use.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with adding rational numbers as constant or coefficients.
- Students may try to combine terms inside of parentheses first that cannot be combined, like rewriting $(x + 2)$ as $2x$.

THINGS TO CONSIDER

- In 9.2.3 Collaboration, discuss the value of the parentheses in adding linear expressions after students model a few problems with algebra tiles. Students should conclude that the parentheses can be dropped and then combine like terms.
- Mathematically, the meaning of “simplest form” is debated, so use caution with how frequently this language is used. It should be varied with language such as “equivalent forms” and “using the fewest terms possible.”

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong

Prompt students to share multiple answers and pathways for other students in 9.2.2 Exploration's Which One Doesn't Belong. There is no one “correct” answer, just like neither of two equivalent expressions is more “correct” in representation (MA.K12.MTR.2.1: Multiple Representations).

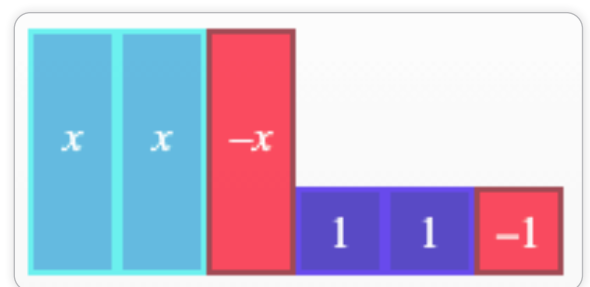
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Students progress from modeling the problems concretely with algebra tiles in 9.2.3 Collaboration to adding linear algebraic expressions with rational coefficients algebraically in the subsequent components of the lesson. Highlight multiple representations throughout the lesson, both in language and in models like algebra tiles, to continue reinforcing concepts of equivalence in multiple representations.

DIFFERENTIATE INSTRUCTION

The use of algebra tiles in 9.2.2 Exploration provides concrete and visual representations of the concepts. Consider using algebra tiles throughout this lesson for students who may struggle with adding like terms and finding the sums, where possible/appropriate, with particular emphasis on zero pairs.



9.2.1 WARM-UP: VISUAL PATTERNS

TEACHER MOVES

MA.K12.MTR.5.1: Patterns and Structure

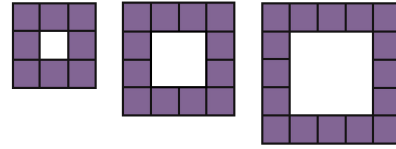
Students who successfully exhibit mathematical thinking and reasoning recognize patterns in what they see and develop generalizations for similar problems. Highlight student moves that embody these qualities.

The patterns and mathematical thinking students exhibit may not exactly match the key but may still be correct mathematical thinking. For example, a student might write the sequence “8, 12, 16” and be able to predict the next figure will have 16 squares. Students do not have to be able to express the pattern algebraically to demonstrate mathematical thinking.



9.2.1 Warm-Up: Visual Patterns

Three figures are shown.



1. What pattern do you see?

Answers will vary.

I see small purple squares making larger squares with a white square in the middle. The square gets bigger every time. The length of the square increases by 1 unit each step in the pattern. The number of purple squares increases by 4 each step.

2. Explain how many small purple squares you think will be in the next figure.

Answers will vary.

The number of small purple squares increases by 4 each time. Since there were 16 small purple squares in the third figure, there will be 20 in the next figure.

9.2.2 EXPLORATION: WHICH ONE DOESN'T BELONG?

DIFFERENTIATE INSTRUCTION

Which One Doesn't Belong

This language routine purposefully provides four different “correct” answers in order to increase entry points and amplify mathematical discussion.

Consider the following possible responses for each expression:

- Expression A: “Only one that begins with a constant” or “Only one that begins with a positive integer.”
- Expression B: “Only one with fractions,” “Only one that begins and ends with a variable,” or “Only one with four terms.”
- Expression C: “Only one with decimals” or “Only one in which the simplified form is $-5x+3$.”
- Expression D: “Only one with only one constant.”



9.2.2 Exploration: Which One Doesn't Belong?

1. Which expression does not belong with the other three? Explain your reasoning.

Answers will vary.

Expression C, because the others are equivalent to $-5x + 4$ while Expression C is equivalent to $-5x + 3$. Expression B because it's the only expression with fractions.

Expression C because it's the only expression with decimals.

Expression A	Expression B
$7 - 5x - 3$ $-5x + 4$	$\frac{3}{2}x + 6 - 2 - 6\frac{1}{2}x$ $-5x + 4$
Expression C	Expression D
$-5x + 7.25 - 4.25$ $-5x + 3$	$-2x - 3x + 4$ $-5x + 4$



9.2.3 Collaboration: Adding Expressions with Algebra Tiles

Algebra tile representations of an expression are given. Work with your partner to find the sum by following the steps below.

1.

On the lines, write the algebraic expression for each representation.	$(2x + 3) + (x + (-5))$
Rearrange the tiles to group like tiles. Draw the representation.	$(2x + x) + (3 + (-5))$
Eliminate any zero pairs to find the sum. Write the sum.	$3x + (-2)$

2.

On the lines, write the algebraic expression for each representation.	$(-3x + 4) + (2x + (-7))$
Rearrange the tiles to group like tiles. Draw the representation.	$((-3x) + 2x) + (4 + (-7))$
Eliminate any zero pairs to find the sum. Write the sum.	$-x + (-3)$

9.2.3 COLLABORATION: ADDING EXPRESSIONS WITH ALGEBRA TILES

DIFFERENTIATE INSTRUCTION

Consider providing algebra tiles for kinesthetic learners interacting with this activity.

Digital algebra tiles are also available in the Student and Teacher Areas.

TEACHER MOVES

Notice the visual representation of the tiles. A common misconception might be the constants interpreted as coefficients. For example, the first set of tiles may be interpreted as $3x^2$ instead of $2x + 3$.

TEACHER MOVES

Notice the use of “zero pairs” when finding the sum.



9.2.4 Guided Instruction: Adding Linear Expressions

When adding two expressions, rearrange the terms so that like terms are grouped together. Then add the coefficients of the like variable terms together and add the constants. Zero pairs are eliminated in this process.



Subtracting an integer is the same as adding the opposite. For example, the expression $9 - 13$ can be rewritten as $9 + -13$.

1. Find the sum of the two expressions.

a.

Original Expression	$(2x + 6) + (5x + 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$2x + 5x + 6 + 4$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$7x + 10$

b.

Original Expression	$(-2x + 6) + (5x - 4)$
Regrouped Expression: Rearrange the terms so that like terms are together.	$-2x + 5x + 6 + (-4)$
Simplified Expression: Add the coefficients of the like variable terms and add the constants.	$3x + 2$

9.2.4 GUIDED INSTRUCTION: ADDING LINEAR EXPRESSIONS

TEACHER MOVES

Consider using language such as, “Rearrange the terms so like terms are together...this is regrouping the expression,” and similarly making connections to the second phrase in each table to “simplifying” so that those connections are clear when students move on to the guided practice with a partner.

TEACHER MOVES

This lesson only prioritizes adding linear expressions, so do not spend time on modeling or explaining the relationship between addition and subtraction. Lesson 3 will cover this.

9.2.5 COLLABORATION: ADDING LINEAR EXPRESSIONS

ENRICHMENT

Notice the prompts with “regrouped” and “simplified” to represent the steps students can take to adding expressions. Use these terms in your question stems as you are monitoring students’ practice.

DIFFERENTIATE INSTRUCTION

Students can also use colors and/or shapes to code like terms when simplifying linear expressions. It may also be helpful to encourage students to circle like terms with different shapes or colors.

TEACHER MOVES

Notice the use of decimals and fractions in parts d and e. Encourage students to either rewrite -5.5 as a fraction or the other fractions as decimals.

QUESTIONING

Lesson 2 is all about adding, while the next lesson begins exploring subtraction. For students who are ready for a challenge, consider priming them with questions like:

- How would the simplified expression be different if the operation between parentheses was subtraction?



9.2.5 Collaboration: Adding Linear Expressions

- Work with your partner to find the sum of the two expressions in simplest form.

a.

Original Expression	$4 + (8 - 0.4x)$
Regrouped Expression	$(-0.4x) + (8 + 4)$
Simplified Expression	$(-0.4x) + 12$

b.

Original Expression	$\left(3\frac{1}{2}x - \frac{3}{5}\right) + \left(2\frac{1}{4}x + \frac{1}{3}\right)$
Regrouped Expression	$\left(3\frac{1}{2}x + 2\frac{1}{4}x\right) + \left(\left(-\frac{3}{5}\right) + \frac{1}{3}\right)$
Simplified Expression	$5\frac{3}{4}x + \left(-\frac{4}{15}\right) = 5\frac{3}{4}x - \frac{4}{15}$

c.

Original Expression	$(-5y + 2) + (5y - 8)$
Regrouped Expression	$((-5y) + 5y) + (2 + (-8))$
Simplified Expression	-6

d.

Original Expression	$(0.7 - 2.4x) + (4.1x - 5.32)$
Regrouped Expression	$((-2.4x) + 4.1x) + (0.7 + (-5.32))$
Simplified Expression	$1.7x + (-4.62) = 1.7x - 4.62$

e.

Original Expression	$\left(\frac{3}{4}w - 5.5\right) + \left(-\frac{1}{2}w + 4\right)$
Regrouped Expression	$\left(\frac{3}{4}w + \left(-\frac{1}{2}w\right)\right) + ((-5.5) + 4)$
Simplified Expression	$\frac{1}{4}w + (-1.5) = \frac{1}{4}w - 1.5$

**9.2.6 Your Turn!**

1. Find the sum of each expression using the fewest terms possible.

a. $(x + 7) + (x - 10)$

$$2x + (-3)$$

$$= 2x - 3$$

b. $(3x - 5) + (5 + 6x)$

$$9x$$

c. $\left(1\frac{6}{7} - y\right) + \left(4y + \frac{1}{7}\right)$

$$3y + 2$$

d. $(4w - 6) + (-6w - 1)$

$$(-2w) + (-7)$$

$$= -2w - 7$$

e. $(9 - 4.6x) + (-5.3x - 7)$

$$-9.9x + 2$$

f. $\left(-\frac{5}{2}k - 7\right) + \left(\frac{1}{5}k + 4\right)$

$$= -\frac{23}{10}k + (-3)$$

$$= -\frac{23}{10}k - 3$$

$$= -2\frac{3}{10}k + (-3)$$

$$= -2\frac{3}{10}k - 3$$

9.2.6 YOUR TURN!**DIFFERENTIATE INSTRUCTION**

Notice the entry points for questions 1a, 1b, and 1d (whole numbers) compared to 1c, 1e, and 1f (fractions and decimals). Consider partitioning practice so that students can choose which set of questions to engage with first.

TEACHER MOVES

Review the sum of each expression before moving on to the Wrap-Up and use this as an opportunity to clarify any lingering misconceptions.

**9.2.7 Wrap-Up: Error Analysis**

1. Your friend finds the sum of $\left(-2\frac{1}{2}x + 7.1\right) + (4x - 5.2)$ to be $-\frac{3}{2}x + 1.9$.

Write a detailed note to your friend explaining why their answer is correct or why it is incorrect.

The answer is incorrect because the sign of the coefficient of x should be positive, not negative.

$$-2\frac{1}{2} + 4 = 1\frac{1}{2} = \frac{3}{2}, \text{ so}$$

$$\left(-2\frac{1}{2}x + 4x\right) + (7.1 + (-5.2)) = \frac{3}{2}x + 1.9.$$

9.2.7 WRAP-UP: ERROR ANALYSIS**QUESTIONING****MA.K12.MTR.4.1: Error Analysis**

Consider highlighting clarification language in MA.K12.MTR.4.1 that students exhibit in their responses.

If student response compares the larger positive value as reasoning behind their analysis, then select, sequence, and present student work to advance and deepen understanding of correct and increasingly efficient methods.

If student response shows evidence of different response using mathematical justification, then develop students' ability to justify methods and compare their responses to the responses of their peers.